



JAVA PROGRAMMING

(340)

STATE 2024

PRODUCTION:

Java

_____ (520 points)

Test Time: 90 minutes

Java

In this test you will be creating another prototype seating chart software for your local school. The program will allow the user to be able to have control options through key board entries. There are three major actions of the program: (1) Randomly create student list; (2) Manually create students; and (3) Exit the program.

Input/Output Option [1]

All user input and output will take place in the console.

STEP 1: The program begins by asking the user what they would like today.

```
Hello. What would you like to do today?
Press [1] to create random students OR Press [2] to create a single student OR Press [3] to end the program
```

STEP 2 Option [1]: After entering 1, the program will ask the user to enter in a value between 1 and 50 (this number is how many students the program will create).

```
Hello. What would you like to do today?
Press [1] to create random students OR Press [2] to create a single student OR Press [3] to end the program
1
Please enter in a value between 1 and 50: [
```

STEP 3 Option [1]: After entering the amount of student records to create, the program will list out the results. In addition, the program also finds the records with the Max GPA and Min GPA. After the list is created, the program returns to STEP 1.

```
Please enter in a value between 1 and 50: 5

1) Mitchell, Jones | Grade Level: 11 | GPA: 3.36
2) Linda, Janet | Grade Level: 12 | GPA: 3.84
3) Turner, Martin | Grade Level: 10 | GPA: 3.4
4) Rose, Johnson | Grade Level: 11 | GPA: 3.05
5) Rose, Irene | Grade Level: 10 | GPA: 2.05

Max GPA: 3.84 Linda, Janet

Min GPA: 2.05 Rose, Irene

Hello. What would you like to do today?
Press [1] to create random students OR Press [2] to create a single student OR Press [3] to end the program
```

Input/Output Option [2]

STEP 1: This the same as before. The user will be entering option [2].

STEPS 2-5 Option [2]: The user will enter in all the student information in this sequence.

```
You are now creating a single student
Enter in the first name
Jose
Enter in the last name
Patel
Enter in the student grade level 9th-12th
9
Enter in the GPA in its proper format (#.##)
3.25
```

- STEP 2: First Name
- STEP 3: Last Name
- STEP 4: Grade Level
- STEP 5: GPA

STEP 6: This is the output for the list. Notice that the record created is added to the end of the list that was previously created (the new student created is the 6th student). The program will still need to find the highest and lowest GPA. After listing, the program returns to STEP 1.

```
1) Mitchell, Jones | Grade Level: 11 | GPA: 3.36
2) Linda, Janet | Grade Level: 12 | GPA: 3.84
3) Turner, Martin | Grade Level: 10 | GPA: 3.4
4) Rose, Johnson | Grade Level: 11 | GPA: 3.05
5) Rose, Irene | Grade Level: 10 | GPA: 2.05
6) Patel, Jose | Grade Level: 9 | GPA: 3.25

Max GPA: 3.84 Linda, Janet

Min GPA: 2.05 Rose, Irene

Hello. What would you like to do today?
Press [1] to create random students OR Press [2] to create a single student OR Press [3] to end the program
```

Input/Output Option [3]

STEP 1: This the same as before. The user will be entering option [3]. This will exit the software.

```
Hello. What would you like to do today?
Press [1] to create random students OR Press [2] to create a single student OR Press [3] to end the program
3
Goodbye!
```

Error Messages:

Input/Output STEP 1 Invalid Data Error

STEP 1: the user attempts to enter invalid data. Error message is given. The program needs to return to STEP 1.

```
Hello. What would you like to do today?
Press [1] to create random students OR Press [2] to create a single student OR Press [3] to end the program
#&$FFF

ERROR: Please enter valid data type.
```

Input/Output STEP 1 Range Violation

STEP 1: the user attempts to enter value outside of the range. “Your entry is out of range.” message is given. The program needs to return to STEP 1.

```
Hello. What would you like to do today?
Press [1] to create random students OR Press [2] to create a single student OR Press [3] to end the program
5
Your entry is out of range.
```

Input/Output STEP 2 Option [1]

Invalid Data Error

STEP 2: the user attempts to enter invalid data. Error message is given. The program needs to return to STEP 2.

```
Please enter in a value between 1 and 50: TTTT
ERROR: Please enter valid data type.
```

Input/Output STEP 2 Option [1] Range Violation

STEP 2: the user attempts to enter value outside of the range. “Your entry is out of range.” message is given. The program needs to return to STEP 2.

```
Please enter in a value between 1 and 50: 444
Your entry is out of range.
```

Input/Output STEP 4 Option [2] Invalid Data Error

STEP 4: the user attempts to enter invalid data. Error message is given. The program needs to return to STEP 4.

```
Enter in the student grade level 9th-12th
w
ERROR: Please enter valid data type.
```

Input/Output STEP 4 Option [2] Range Violation

STEP 4: the user attempts to enter value outside of the range. “Your entry is out of range.” message is given. The program needs to return to STEP 4.

```
Enter in the student grade level 9th-12th
222
Your entry is out of range.
```

Input/Output STEP 5 Option [2]

Invalid Data Error

STEP 5: the user attempts to enter invalid data. Error message is given. The program needs to return to STEP 5.

```
Enter in the GPA in its proper format (##)
www
ERROR: Please enter valid data type.
```

Input/Output STEP 5 Option [2] Range Violation

STEP 5: the user attempts to enter value outside of the range. “Your entry is out of range.” message is given. The program needs to return to STEP 5.

```
Enter in the GPA in its proper format (##)
222
Your entry is out of range.
```

Solution and Project (There is NO partial credit) (NOTE: UC represents uppercase and LC represents lowercase)		
The Java source file is present on the flash drive in a single folder with your contest ID		20 points
Program Execution (If the program does not execute, then the remaining items in this section receive a score of zero)		
Program starts STEP 1		10 points
STEP 1 Invalid Data Error is handled appropriately		20 points
STEP 1 Range Violation is handled appropriately		10 points
Option [1] is selected, program moves to appropriate STEP 2		10 points
STEP 2 Option [1] Invalid Data Error is handled appropriately		20 points
STEP 2 Option [1] Range Violation is handled appropriately		10 points
STEP 3 Option [1] Random students matches output & correct Max/Min GPA displayed		30 points
After STEP 3 Option [1] Program returns immediately to active STEP 1		20 points
Option [2] is selected, program moves to appropriate STEP 2		10 points
STEP 4 Option [2] (Grade Level) Invalid Data Error is handled appropriately		20 points
STEP 4 Option [2] (Grade Level) Range Violation is handled appropriately		10 points
STEP 5 Option [2] (GPA) Invalid Data Error is handled appropriately		20 points
STEP 5 Option [2] (GPA) Range Violation is handled appropriately		10 points
STEP 6 Option [2] Student created in STEPS 2-5 appears in auto generated student original formatted list with proper formatting (first letter of names UC; Grade level is whole number; GPA #.##)		30 points
After STEP 6 Option [2] Program returns immediately to active STEP 1		20 points
STEP 1 Option [3] program exits with goodbye message.		10 points
		20 points
Subtotal		/280 Points

Source Code Review (There is NO partial credit) <i>NOTE: you must place the comment flag in front of the comment in your code in order to get credit. The comment flag will precede the explanation. For example, if the flag is SC1, your comment must read as “#SC1...” in front of the part of the code being reviewed. Code must work to get credit. There is no partial credit.</i>		
A comment containing the contestant number is present at the top of the SeatingChartBuilderState.java file		10 points
SC1: Students.java class: All constructors, methods, and variables are appropriately created; no partial credit		40 points
SC2: SeatingChartBuilderState class: create Scanner and ArrayList objects (place SC2 by both instances); no partial credit		10 points
SC3: SeatingChartBuilderState class <i>printHighestGPA()</i> : code that compares GPA's for all students		20 points
SC4: SeatingChartBuilderState class <i>printLowestGPA()</i> : code that compares GPA's for all students		20 points
SC5: SeatingChartBuilderState class <i>inputManager()</i> : code with try/catch exception for invalid data error and range violation of STEP 1 (place SC5 by both instances)		20 points
SC6: SeatingChartBuilderState class <i>inputManager()</i> : code with try/catch exception for invalid data error and range violation of STEP 2 Option[1] (place SC6 by both instances)		20 points
SC7: SeatingChartBuilderState class <i>inputManager()</i> : code with try/catch exception for invalid data error and range violation of STEP 4 Option[2] (place SC7 by both instances)		20 points
SC8: SeatingChartBuilderState class <i>inputManager()</i> : code with try/catch exception for invalid data error and range violation of STEP 5 Option[2] (place SC8 by both instances)		20 points
SC9: SeatingChartBuilderState class <i>inputManager()</i> : code that formats GPA with proper decimal format object (place SC9 by all uses of this object)		10 points
SC10: SeatingChartBuilderState class <i>inputManager()</i> : code that formats first and last name to required convention (first letter is UC, all other letters are LC)		20 points
SC11: SeatingChartBuilderState class <i>inputManager()</i> : code that exits the program STEP 1 Option [3]		10 points
	Subtotal	/240 Points
	Total Points	/520 Points

```

1
2 import java.io.FileNotFoundException;
3 import java.io.File;
4 import java.util.*;
5 import java.text.DecimalFormat;
6
7 public class SeatingChartBuilderState
8 {
9
10     static Scanner sc = new Scanner(System.in); //STATE POINT //SC2
11     static ArrayList<Students> students = new ArrayList<Students>();//STATE POINT //SC2
12     public static void main (String args [])
13     {
14
15         inputManager(); //Given
16
17         printStudents();
18         printHighestGPA(); //Given
19         printLowestGPA(); //Given
20
21     }
22
23
24
25     //STATE POINT
26     //Students will create. Signature only
27     private static void printHighestGPA() //ArrayList<Students> stuList
28     {
29         int ind = 0;
30         double num = 0;
31
32         for(int i = 1; i < students.size(); i++)
33         {
34             if(students.get(i).getGPA()>students.get(ind).getGPA()) //SC3
35             {
36                 ind = i;
37                 num = students.get(i).getGPA();
38             }
39         }
40
41         System.out.println("\n\nMax GPA: "+ students.get(ind).getGPA()+ " "+students.get(ind).getWholeName())
42 ;
43     }
44
45     //STATE POINT
46     //Students will create. Signature only
47     private static void printLowestGPA()
48     {
49         int ind = 0;
50         double num = 0;
51
52         for(int i = 1; i < students.size(); i++)
53         {
54             if(students.get(i).getGPA()<students.get(ind).getGPA()) //SC4
55             {
56                 ind = i;
57                 num = students.get(i).getGPA();
58             }
59         }
60
61         System.out.println("\n\nMin GPA: "+ students.get(ind).getGPA()+ " "+students.get(ind).getWholeName())
62 ;
63     }
64

```

```

65     //Given
66     private static void setStudents(int nc)
67     {
68         Random rand = new Random();
69         Students s;
70         DecimalFormat df = new DecimalFormat("#.00");
71         int nameCount = nc;
72         double gpa;
73         int grade;
74         String fn; String ln;
75
76         String [] allNames ={"Walter","Jones","Rose","Wilson" ,"Jack", "Rodriguez" , "Elizabeth" , "Smith", "
Earl", "Carter", "Linda", "Ward", "Christopher",
77         "Turner", "Martin", "Murphy", "Betty", "Garcia", "Shawn", "Taylor","Sean", "Simmons", "Joshua", "E
vans", "Norma", "Mitchell", "Brenda", "Johnson", "Donna",
78         "Clark", "Irene", "Diaz","Marilyn", "Coleman","Arthur", "Collins","Henry", "Hall","Howard", "Robin
son","Jerry", "Green","Maria", "Price", "Evelyn", "Bell",
79         "Janet", "Moore", "Susan", "Foster"};
80
81         for(int i =0; i<nameCount; i++)
82         {
83
84             do{
85
86                 fn = allNames[rand.nextInt(allNames.length)];
87                 ln = allNames[rand.nextInt(allNames.length)];
88                 }while(fn.equals(ln));
89                 grade = rand.nextInt(4)+9;
90                 double tempGPA = rand.nextDouble()*3.0 + 1.0;
91                 String tempString =df.format(tempGPA);
92                 gpa = Double.parseDouble(tempString);
93                 s = new Students(fn,ln,grade,gpa);
94                 students.add(s);
95             }
96         }
97
98
99     //Given
100    private static void printStudents(){
101        int i = 1;
102        for (Students n : students)
103        {
104            System.out.println(i+" "+n.getWholeName() + " | Grade Level: "+n.getGradeLevel()+" | GPA: "+n.get
GPA());
105            i++;
106        }
107
108        printHighestGPA();
109        printLowestGPA();
110    }
111
112
113    private static void inputManager()
114    {
115        int temp =0;
116        int grade =0;
117        double GPA =0.0;
118        String firstName;
119        String lastName;
120        String tempString;
121        Students s;
122        DecimalFormat df = new DecimalFormat("#.00"); //SC9
123
124
125        while(true){
126            try{ //SC5

```

```

127         do{
128             temp =0;
129             System.out.println("_____")
130         };
131         System.out.println("Hello. What would you like to do today?");
132         System.out.println("Press [1] to create random students OR Press [2] to create a single student OR Press [3] to end the program");
133         temp = sc.nextInt();
134         if(temp <= 0 || temp > 3) System.out.println("Your entry is out of range."); //SC5
135         sc.nextLine();
136         } while(temp <= 0 || temp > 3);
137         if(temp ==1){
138             do{
139                 temp =0;
140                 try{ //SC6
141                     do{
142                         System.out.print("Please enter in a value between 1 and 50: ");
143                         temp = sc.nextInt();
144                         if(temp >50 || temp <1) System.out.println("Your entry is out of range."); //SC6
145                         sc.nextLine();
146                     }while(temp>50|| temp<1);
147                     catch(InputMismatchException e) //SC6
148                     {
149                         sc.next();
150                         System.out.println("\nERROR: Please enter valid data type.");
151                     }
152                     } while(temp >50 || temp <1);
153                 setStudents(temp);
154                 temp =0;
155                 System.out.println("_____")
156             };
157             printStudents();
158         }
159         if(temp ==2){
160             System.out.println("You are now creating a single student");
161             System.out.println("Enter in the first name");
162             firstName = sc.nextLine().toLowerCase();
163
164             System.out.println("Enter in the last name");
165             lastName = sc.nextLine().toLowerCase();
166             do{
167                 try{ //SC7
168                     do{
169                         System.out.println("Enter in the student grade level 9th-12th");
170                         grade = sc.nextInt();
171                         if(grade >12 || grade <9) System.out.println("Your entry is out of range."); //SC7
172                         }while(grade >12 || grade <9);
173                     }
174                     catch(InputMismatchException e) //SC7
175                     {
176                         sc.next();
177                         System.out.println("\nERROR: Please enter valid data type.");
178                     }
179                     }while(grade >12 || grade <9);
180                 do{
181                     try{ //SC8
182                         do{
183                             System.out.println("Enter in the GPA in its proper format (##.##)");
184                             GPA = sc.nextDouble();
185                             tempString =df.format(GPA); //SC9
186                             GPA = Double.parseDouble(tempString);
187                             if(GPA>4.0 || GPA <= 0.0) System.out.println("Your entry is out of range."); //SC8
188                             }while(GPA>4.0 || GPA <= 0.0);
189                         }

```

```

190         catch(InputMismatchException e) //SC8
191         {
192             sc.next();
193             System.out.println("\nERROR: Please enter valid data type.");
194         }
195         }while(GPA>4.0 || GPA <= 0.0);
196         firstName = firstName.substring(0,1).toUpperCase() + firstName.substring(1).toLowerCase(); /
/SC10
197         lastName = lastName.substring(0,1).toUpperCase() + lastName.substring(1).toLowerCase(); //SC
10
198         s = new Students(firstName,lastName,grade,GPA);
199         students.add(s);
200         System.out.println("_____")
;
201         printStudents();
202     }
203     if(temp ==3){
204         System.out.println("Goodbye!");
205         System.exit(0); //SC11
206     }
207 }
208 catch(InputMismatchException e) //SC5
209 {
210     sc.next();
211     System.out.println("\nERROR: Please enter valid data type.");
212 }
213 }
214
215 }
216
217
218 }
219
220 //SC1
221 class Students
222 {
223     //Create appropriately variables based upon constructor parameters
224     String first_Name;
225     String last_Name;
226     int grade_Level;
227     double GPA;
228     //Generic Constructor
229     public Students()
230     {
231         first_Name = "Dee";
232         last_Name = "Fault";
233         grade_Level = 0;
234         GPA = 0.0;
235     }
236     //Constructor with Parameters
237     public Students(String fn, String ln, int gl, double gpa)
238     {
239         first_Name = fn;
240         last_Name = ln;
241         grade_Level = gl;
242         GPA = gpa;
243     }
244
245     //Returns "last name, first name" with appropriate variables
246     public String getWholeName()
247     {
248         return last_Name + ", "+first_Name;
249     }
250     //Returns first name
251     public String getFirstName()
252     {

```

```
253     return first_Name;
254     }
255     //Returns last name
256     public String getLastName()
257     {
258         return last_Name;
259     }
260     //Returns the GPA
261     public double getGPA()
262     {
263         return GPA;
264     }
265     //Returns grade level
266     public int getGradeLevel()
267     {
268         return grade_Level;
269     }
270
271
272 }
```